Hyeon Jeon

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Education

2022 – Present **Ph.D. Student in Computer Science and Engineering,** Seoul National University
Thesis: Dimensionality Reduction Considered Harmful (Some of the Time)
Committee: Jinwook Seo (Advisor and Vice Chair), Jaesik Park (Chair),
Ghulam Jilani Quadri, Michaël Aupetit, and Kwan-Liu Ma

2020 – 2022 **Master's Student in Computer Science and Engineering,** Seoul National University
Advisor: Jinwook Seo
Entered Ph.D. program in Feb. 2022

2016 – 2020 **Bachelor of Science in Computer Science and Engineering,** POSTECH
Graduate with honors (Magna Cum Laude)
Early graduation (Seven semesters)

Research Interests

Hyeon works at the intersection of data visualization, machine learning, and human-computer interaction. He aims to make visual analytics and communications more reliable.

Publications

Preprints

2025 P3 **Stop misusing t-SNE and UMAP for visual analytics**
Hyeon Jeon, Jeongin Park, Sungbok Shin, and Jinwook Seo
arXiv:2506.08725

P2 **Data therapist: Eliciting domain knowledge from subject matter experts using large language models**
Sungbok Shin, **Hyeon Jeon**, Sanghyun Hong, and Niklas Elmqvist
arXiv:2505.00455

P1 **Reading.help: Supporting EFL readers with proactive and on-demand explanation of english grammar and semantics**
Sunghyo Chung, **Hyeon Jeon**, Sungbok Shin, and Md Naimul Hoque
arXiv:2505.14031

Peer-Reviewed Conference Full Papers & Journal Articles

Papers in the IEEE VIS proceedings are published as a special issue of TVCG.
Papers published in TVCG are invited for presentation at related conferences like VIS.
* denotes equal contribution.

2025 J15 **Distortion-aware brushing for reliable cluster analysis in multidimensional projections**
Hyeon Jeon, Michaël Aupetit, Soohyun Lee, Kwon Ko, Youngtaek Kim, Ghulam Jilani Quadri, and Jinwook Seo
IEEE Transactions on Visualization and Computer Graphics (TVCG), under minor revision

J14 **UMATO: Bridging local and global structures for reliable visual analytics with dimensionality reduction**
Hyeon Jeon, Kwon Ko, Soohyun Lee, Jake Hyun, Taehyun Yang, Gyehun Go, Jaemin Jo, and Jinwook Seo
IEEE Transactions on Visualization and Computer Graphics (TVCG)

J13 **Dataset-adaptive dimensionality reduction**
Hyeon Jeon, Jeongin Park, Soohyun Lee, Dae Hyun Kim, Sungbok Shin, and Jinwook Seo
IEEE Transactions on Visualization and Computer Graphics (TVCG, Proc. VIS 2025)

J12 **A critical analysis of the usage of dimensionality reduction in four domains**
Dylan Cashman, Mark Keller, **Hyeon Jeon**, Bum Chul Kwon, and Qianwen Wang
IEEE Transactions on Visualization and Computer Graphics (TVCG)
Invited to IEEE VIS 2025 for presentation

J11 **Measuring the validity of clustering validation datasets**
Hyeon Jeon, Michaël Aupetit, DongHwa Shin, Aeri Cho, Seokhyeon Park, and Jinwook Seo
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)

J10 **Unveiling high-dimensional backstage: A survey for reliable visual analytics with dimensionality reduction**
Hyeon Jeon, Hyunwook Lee, Yun-Hsin Kuo, Taehyun Yang, Daniel Archambault, Sungahn Ko, Takanori Fujiwara, Kwan-Liu Ma, and Jinwook Seo
ACM Conference on Human Factors in Computing Systems (CHI 2025)

2024 J9 **Phenoflow: A human-llm driven visual analytics system for exploring large and complex stroke datasets**
Jaeyoung Kim, Sihyeon Lee, **Hyeon Jeon**, Keon-Joo Lee, Hee-Joon Bae, Bohyoung Kim, and Jinwook Seo
IEEE Transactions on Visualization and Computer Graphics (TVCG, Proc. VIS 2024)

J8 **Natural language dataset generation framework for visualizations powered by large language models**
Hyung-Kwon Ko, **Hyeon Jeon**, Gwanmo Park, Dae Hyun Kim, Nam Wook Kim, Juho Kim, and Jinwook Seo
ACM Conference on Human Factors in Computing Systems (CHI 2024)

J7 **CloChat: Understanding how people customize, interact, and experience personas in large language models**
Juhye Ha, **Hyeon Jeon**, DaEun Han, Jinwook Seo, and Changhoon Oh
ACM Conference on Human Factors in Computing Systems (CHI 2024)

2023 J6 **MoNetExplorer: A visual analytics system for analyzing dynamic networks with temporal network motifs**
Seokweon Jung, DongHwa Shin, **Hyeon Jeon**, Kiroong Choe, and Jinwook Seo
IEEE Transactions on Visualization and Computer Graphics (TVCG)
Invited to IEEE VIS 2024 for presentation

J5 **Classes are not clusters: Improving label-based evaluation of dimensionality reduction**
Hyeon Jeon, Yun-Hsin Kuo, Michael Aupetit, Kwan-Liu Ma, and Jinwook Seo
IEEE Transactions on Visualization and Computer Graphics (TVCG, Proc. VIS 2023)

J4 **CLAMS: A cluster ambiguity measure for estimating perceptual variability in visual clustering**
Hyeon Jeon*, Ghulam Jilani Quadri*, Hyunwook Lee, Paul Rosen, Danielle Albers Szafir, and Jinwook Seo
IEEE Transactions on Visualization and Computer Graphics (TVCG, Proc. VIS 2023)
Best Papers Honorable Mention Awards (top 5% submissions)

J3 **Large-scale text-to-image generation models for visual artists' creative works**
ACM International Conference on Intelligent User Interfaces (IUI 2023)
Hyung-Kwon Ko, Gwanmo Park, **Hyeon Jeon**, Jaemin Jo, Juho Kim, and Jinwook Seo

2021 J2 **Measuring and explaining the inter-cluster reliability of multidimensional projections**
IEEE Transactions on Visualization and Computer Graphics (TVCG, Proc. VIS 2021)
Hyeon Jeon, Hyung-Kwon Ko, Jaemin Jo, Youngtaek Kim, and Jinwook Seo

2020 J1 **Githru: Visual analytics for understanding software development history through git metadata analysis**
IEEE Transactions on Visualization and Computer Graphics (TVCG, Proc. VAST 2020)
Youngtaek Kim, Jaeyoung Kim, **Hyeon Jeon**, Young-Ho Kim, Hyunjoo Song, Bohyoung Kim, and Jinwook Seo

Peer-Reviewed Conference Short Papers

2025 S9 **Metric design != metric behavior: Improving metric selection for the unbiased evaluation of dimensionality reduction**
Jiyeon Bae, **Hyeon Jeon**, and Jinwook Seo
IEEE Visualization and Visual Analytics (VIS 2025)

S8 **Navigating high-dimensional backstage: A guide for exploring literature for the reliable use of dimensionality reduction**
Hyeon Jeon, Hyunwook Lee, Yun-Hsin Kuo, Taehyun Yang, Daniel Archambault, Sungahn Ko, Takanori Fujiwara, Kwan-Liu Ma, and Jinwook Seo
Eurographics/VGTC Conference on Visualization (EuroVis 2025)

S7 **Automated pipeline for detecting and analyzing misleading visual elements**
Min Hyeong Kim, Yumin Song, Yungun Kim, Aeri Cho, Soohyun Lee, **Hyeon Jeon**, and Jinwook Seo
IEEE Pacific Visualization Conference (PacificVis 2025)

2024 S6 **CLeVer: Continual learning visualizer for detecting task transition failure**
Minsuk Chang, Donghun Kim, **Hyeon Jeon**, Seokweon Jung, and Jinwook Seo
IEEE Pacific Visualization Conference (PacificVis 2024)

S5 **IoLens: Visual analytics system for exploring storage i/o tracking process**
Changmin Jeon, Jiwon Ha, Hyolim Hong, **Hyeon Jeon**, Hyeonsang Eom, Heonyoung Yeom, and Jinwook Seo
IEEE Pacific Visualization Conference (PacificVis 2024)

2023 S4 **ZADU: A python library for evaluating the reliability of dimensionality reduction embeddings**
Hyeon Jeon, Aeri Cho, Jinhwa Jang, Soohyun Lee, Jake Hyun, Hyung-Kwon Ko, Jaemin Jo, and Jinwook Seo
IEEE Visualization and Visual Analytics (VIS 2023)

2022 S3 **Uniform manifold approximation with two-phase optimization**
Hyeon Jeon*, Hyung-Kwon Ko*, Soohyun Lee, Jaemin Jo, and Jinwook Seo
IEEE Visualization and Visual Analytics (VIS 2022)

S2 **VANT: A visual analytics system for refining parallel corpora in neural machine translation**
Sebeom Park, Soohyun Lee, Youngtaek Kim, **Hyeon Jeon**, Seokweon Jung, Jinwook Bok, and Jinwook Seo
IEEE Pacific Visualization Symposium (PacificVis 2022)
Best Visualization Notes (top 1st submission)

2021 S1 **Visualization support for multi-criteria decision making in software issue propagation**
Youngtaek Kim, **Hyeon Jeon**, Young-Ho Kim, Yuhoon Ki, Hyunjoo Song, and Jinwook Seo
IEEE Pacific Visualization Symposium (PacificVis 2021)

Panel & Tutorial Organizations

2025 A2 **Reliable visual analytics with dimensionality reduction: Quality evaluation and interpretation of projections**
Hyeon Jeon, Takanori Fujiwara, and Rafael M. Martins
Tutorial at EuroVis 2025

2024 A1 **How many evaluations are enough? A panel discussion on evaluation trends in information visualization**
Ghulam Jilani Quadri, Danielle Albers Szafir, Arran Zeyu Wang, and **Hyeon Jeon**
Panel at IEEE VIS 2024
Panelists: Niklas Elmqvist, Tobias Isenberg, Rita Borgo, Michael Sedlmair, Cindy Xiong Bearfield

Workshop Papers

2025 W3 **Changing the paradigm from dynamic queries to LLM-generated SQL queries with human intervention**
Assor Ambre, **Hyeon Jeon**, Sungbok Shin, and Jean-Daniel Fekete
Workshop on Visual Analytics in Healthcare (VAHC) at IEEE VIS

W2 **Understanding bias in perceiving dimensionality reduction projections**
Seoyoung Doh, **Hyeon Jeon**, Sungbok Shin, Ghulam Jilani Quadri, Nam Wook Kim, and Jinwook Seo
VisComm 2025 Workshop at IEEE VIS

2023 W1 **A vega-lite dataset and natural language generation pipeline with large language models**
Hyung-Kwon Ko, **Hyeon Jeon**, Gwanmo Park, Dae Hyun Kim, Nam Wook Kim, Juho Kim, and Jinwook Seo
NLVIZ 2023 Workshop at IEEE VIS

Posters and Domestics

2025 O7 **Can VLMs assess similarity between graph visualizations?**
Seokweon Jung, **Hyeon Jeon**, Jeongmin Rhee, and Jinwook Seo
IEEE Pacific Visualization Conference (PacificVis 2025), Poster
Best Poster Awards (top 1st submission)

O6 **Understanding how visualization researchers create visualizations**
Yumin Song, **Hyeon Jeon**, John Joon Young Chung, and Jinwook Seo
IEEE Pacific Visualization Conference (PacificVis 2025), Poster

O5 **HookLens: Visual analytics for understanding react hook structures**
Suyeon Hwang, Minkyu Kweon, Jeongmin Rhee, Soohyun Lee, Seokhyeon Park, Seokweon Jung, **Hyeon Jeon**, and Jinwook Seo
IEEE Pacific Visualization Conference (PacificVis 2025), Poster

2024 O4 **Efficiently crowdsourcing visual importance with punch-hole annotation**
Minsuk Chang, Soohyun Lee, Aeri Cho, **Hyeon Jeon**, Seokhyeon Park, Cindy Xiong Bearfield, and Jinwook Seo
IEEE Visualization and Visual Analytics (VIS 2024), Poster

O3 **Combinational nonuniform timeslicing of dynamic networks**
Seokweon Jung, DongHwa Shin, **Hyeon Jeon**, and Jinwook Seo
IEEE Pacific Visualization Conference (PacificVis 2024), Poster

2023 O2 **Interactive visual analytics system for criminal intelligence analysts with multiple coordinated views**
Seokweon Jung, DongHwa Shin, Jinwook Bok, Seokhyeon Park, **Hyeon Jeon**, Jinwook Seo, Insoo Lee, and Sooyoung Park
Journal of KIISE

2021 O1 **Interactive visualization for exploring information fragments in software repositories**
Youngtaek Kim, **Hyeon Jeon**, Kiroong Choe, Hyunjoo Song, Bohyoung Kim, Jinwook Seo
IEEE Pacific Visualization Symposium (PacificVis 2021), Poster

Awards and Honors

Paper Awards

2025 **Best Poster Awards,** IEEE PacificVis 2025
Top 1st submission among all poster submissions

2023 **Best Papers Honorable Mentions,** IEEE VIS 2023
Top 5% submissions among all full paper submissions

2022 **Best Visualization Notes,** IEEE PacificVis 2022
Top 1st submission among all short paper submissions

Fellowships and Scholarships

2025 **International Collaborative Research Scholarship,** SNU BK21 Four
Granted 5,500,000 KRW (≃ 4,000 USD)

2024 **Google PhD Fellowship,** Google
Granted 10,000 USD

2024 **Future Innovation Individual (Gold Award),** SNU BK21 Four
1st place among student researchers participating in SNU BK21 Four
Granted 3,000,000 KRW (≃ 2,700 USD)

2023 **Outstanding Research Individual Fellowship,** SNU BK21 Four, Dept. of CSE
Granted 10,000,000 KRW (≃ 9,000 USD)

2023 **Star Student Researcher Award,** SNU BK21 Four, Dept. of CSE
Granted 700,000 KRW (≃ 600 USD)

2016 – 2019 **Jigok Scholarship,** POSTECH
Full scholarship for undergraduates (Seven semesters)

Travel Grants

2024 **Conference Travel Award,** Korea Computer Congress
Support for conference registration and travel expenses

Acknowledgements for Community Services

2024 – 2025 **Special Recognitions for Outstanding Reviews**
PacificVis 2024, PacificVis 2025, C&C 2025, CHI 2025 (×2)

2025 **Outstanding Reviewer**
KDD 2025

Academic Services

Program Committee

2026 Full Papers, ACM Conference on Intelligent User Interfaces (IUI)
2025 Short Papers, IEEE Visualization and Visual Analytics (VIS)
2025 Late-Breaking Work, ACM CHI Conference on Human Factors in Computing Systems (CHI)
2024 Video Showcase, ACM CHI Conference on Human Factors in Computing Systems (CHI)

Organizing Committee

2024 – 2025 **Student Volunteer Chair**, IEEE Visualization and Visual Analytics (VIS)
2024 – 2025 **Organizer**, Korea Visualization Workshop (K-VIS)

Student Volunteers

2022 – 2024 IEEE Visualization and Visual Analytics (VIS)
2023 IEEE Pacific Visualization Symposium (PacificVis)
2022 ACM CHI Conference on Human Factors in Computing Systems (CHI)

Conference Session Chair

2025 ACM CHI Conference on Human Factors in Computing Systems (CHI)

Conference Reviewer (External)

2022 – 2025 IEEE Visualization and Visual Analytics (VIS)
2022 – 2026 IEEE Pacific Visualization Symposium (PacificVis)
2022 – 2025 Eurographics/VGTC Conference on Visualization (EuroVis)
2022 – 2025 ACM SIGCHI Conference on Human Factors in Computing Systems (CHI)
2024 – 2025 ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)
2025 ACM Symposium on User Interface Software and Technology (UIST)
2025 ACM Creativity and Cognition Conference (C&C)
2025 ACM Conference on Designing Interactive Systems (DIS)
2024 ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)
2023 ACM Conference on Intelligent User Interfaces (IUI)
2022 Conference on Neural Information Processing System (NeurIPS)
2025 Korea Computer Congress (KCC)

Invited Journal Reviewer

2024 – 2025 IEEE Transactions on Visualizations and Computer Graphics (TVCG)
2025 Computer Graphics Forum
2023 – 2024 Journal of Visualization and Interaction (JoVI)
2025 IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
2025 Springer Machine Learning
2025 Scientific Reports
2024 IEEE Transactions on Geoscience and Remote Sensing (TGRS)
2024 International Journal of Human-Computer Interaction (IJHCI)
2023 ACM Transactions on Knowledge Discovery from Data (TKDD)
2021 – 2025 Visual Informatics

Funded Grants

2024 – 2026 **GuardVis: AI-Powered Validation for Secure and Reliable Data Analysis and Visualization**
National Research Foundation of Korea, Ph.D. Research Fellowship, No. RS-2024-00409716
Investigator: **Hyeon Jeon**
Amount: 40,000,000 KRW (≃ 30,000 USD)
Accepted, but withdrawn due to my alternative military service as a technical research personnel

Experiences

Research Experiences

2025 **Visiting Ph.D. Student**, Aviz, INRIA (advised by Dr. Jean-Daniel Fekete and Dr. Petra Isenberg)
2022 **Visiting Ph.D. Student**, VIDi, University of California, Davis (advised by Dr. Kwan-Liu Ma)
2020 **Undergraduate Intern**, Human-Computer Interaction Lab., Seoul National University

Teaching Experiences

Instructor

2025 **Reliable Visual Analytics with Dimensionality Reduction,** Tutorial at EuroVis 2025 [A2]
Designed the first half of the lecture and programming practice (DR evaluation)
ZADU [S4] was used for the programming practice
Co-instructed with Taknori Fujiwara and Rafael M. Martins

2025 **Introduction to HCI Research,** Seoul National University
Designed the course materials and assignments for a six-week intensive course
Instructed for both the HCI Lab (CSE department) and the ViBA Lab (DS department)

Teaching Assistants

2021 – 2025 **[M1522.000500] Information Visualization and Visual Analytics,** Seoul National University
Contributed to course materials and programming assignments
Advised four team projects to be conference papers [S2, S5, S6, S7]

2021 **[035.001] Digital Computer Concept and Practice,** Seoul National University
Contributed to course materials and programming assignments (Python)

2025 **[E31.103] Undergraduate Independent Research: Inquiry,** Seoul National University
Advised an undergraduate research project on HCI

2025 **AI & Data Science Special Lecture for DS Division,** Samsung Electronics
Course title: Trustworthy multidimensional visualizations
Designed the course syllabus and programming practice session (Python)
ZADU [S4] was used for the programming practice

2023 **Samsung AI Expert Program,** Seoul National University
Course title: Visual analytics for high-dimensional data
Designed the course syllabus and programming practice session (Python)
ZADU [S4] was used for the programming practice

2020 – 2021 **Samsung DS^2 Program,** Seoul National University
Course title: Data Visualization
Contributed to course materials and programming assignments (Vega-Altair)

Tutoring and Mentoring

2021 – 2025 **Undergraduate Research Opportunities Program (UROP),** Seoul National University
Mentored three undergraduate research projects (2021, 2023, and 2025)

2019 – 2020 **CSE Student Mentoring Program (SMP),** POSTECH
Algorithms (CSED331), Automata & Formal Languages (CSED341)

Research Mentoring with Artifacts

All individuals listed have consented to be named in my CV.

2025 **Seoyoung Doh,** CSE & Business Undergraduate Student, Seoul National University
Co-authored [W2]

2024 – 2025 **Jeongin Park,** CSE Undergraduate Student, Seoul National University
Co-authored [P3], [J13], then a Ph.D. student at Seoul National University

2023 – 2025 **Min Hyeong Kim,** CSE Undergraduate/MS Student, Seoul National University
Co-authored [S7], then a Ph.D. student at Seoul National University

2023 – 2025 **Taehyun Yang,** CSE Undergraduate Student, Seoul National University
Co-authored [J10], [S8], then a Ph.D. student at the University of Maryland

2023 – 2024 **Jinhwa Jang,** CSE Master's Student, Seoul National University
Co-authored [S4], then an engineer at Samsung Electronics

2023 **Minsuk Chang,** CSE Undergraduate Student, Seoul National University
Co-authored [S6], [O4], then a Ph.D. student at Georgia Institute of Technology

2021 – 2023 **Soohyun Lee,** CSE Master's Student, Seoul National University
Co-authored [J13–15], [S2–4], [S7], then a Ph.D. student at Seoul National University

2022 **Sebeom Park,** CSE Master's Student, Seoul National University
Co-authored [S2], then a staff engineer at Samsung Electronics

Invited Talks

2025 **Dimensionality reduction considered harmful (some of the time)**
Inha University (Guest Lecture, Recommender Systems [ECE7108]), October 2025
Seoul National University (Graduate School of Data Science), June 2025
University of Stuttgart (VISUS), June 2025
Technical University of Munich, June 2025

2025 **Human-centered dimensionality reduction**
Aarhus University (CA^3), June 2025
Yonsei University (axlab), March 2025
University of Oklahoma (Guest Lecture, Information Visualization [CS5970]), February 2025
Seoul National University, AI / Computing Frontier Winter School, January 2025

2024 – 2025 **Towards more reliable visual analytics for high-dimensional data**
Samsung AI & Data Science Special Lecture, January 2025
KAIST (Interactive Computing Lab), September 2024
POSTECH (HIS Lab), July 2024
University of California, Davis (Visualization Research Group), May 2024
Georgia Institute of Technology (Visualization Group), April 2024
KAIST (Kixlab), February 2024

Google, February 2024
Carnegie Mellon University (Data Interaction Group), February 2024

2022 – 2023 **Making clusters more reliable**
Sungkyunkwan University (Interactive Data Computing Lab), August 2023
UNIST (HAiV Lab Visualization Seminar 2023), August 2023
Sejong University (Data Visualization Lab), May 2023
University of North Carolina, Chapel Hill (Dept. of Computer Science), November 2022

2024 **Ambiguity in data visualizations**
UNIST (HAiV Lab Visualization Seminar 2024) July 2024
UNIST (Guest Lecture, Introduction to Human-Computer Interaction [CSE333]), May 2024

2024 **Being an HCI Researcher**
Seoul National University, HCI & AI Symposium (College of Liberal Studies), October 2024

2025 **Dataset-adaptive dimensionality reduction**
HKUST & SNU Joint HCI+VIS Workshop, July 2025

2024 – 2025 **Stop misusing t-SNE and UMAP for visual analytics**
Korea Visualization Workshop, July 2025
Korea Visualization Workshop, October 2024

2025 **Unveiling high-dimensional backstage: A survey for reliable visual analytics with dimensionality reduction**
Korea Computer Congress, June 2025

2024 **Classes are not clusters: Improving label-based evaluation of dimensionality reduction**
Korea Computer Congress, June 2024

2023 **CLAMS: A cluster ambiguity measure for estimating perceptual variability in visual clustering**
Korea Software Congress, December 2023
Seoul National University, Computing Frontier Summer School, August 2023

2022 **Measuring and explaining the inter-cluster reliability of multidimensional projections**
Korea Software Congress, December 2022

Press Coverage

2024 **Announcing the 2024 Google PhD Fellows**
Google Korea Blog (Korean)

2023 **NAVER-SNU joint research, honored as 'Top 5%' at the prestigious data analysis conference**
The Electronic Times (Korean)

2022 **Measuring and explaining the inter-cluster reliability of multidimensional projections**
Seoul National University Artificial Intelligence Institute (SNU AIIS) Tech Blog

References

Dr. Jinwook Seo, Professor, Seoul National University
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Dr. Kwan-Liu Ma, Distinguished Professor, University of California, Davis
klma@ucdavis.edu

Dr. Michaël Aupetit, Senior Scientist, QCRI, Hamad Bin Khalifa Univiversity
maupetit@hbku.edu.qa

Dr. Ghulam Jilani Quadri, Assistant Professor, University of Oklahoma
quadri@ou.edu

Dr. Jaemin Jo, Associate Professor, Syunkyunkwan University
jmjo@g.skku.edu